

AI & Machine Learning – Task 2 Report

Project: Feature Engineering, Model Optimization & Performance Comparison

Objective

The objective of this project is to compare multiple machine learning regression models using the California Housing Dataset and identify the best-performing model based on RMSE and R^2 Score.

Methodology

1. Loaded the California Housing Dataset.
2. Performed data preprocessing and feature inspection.
3. Applied StandardScaler for feature scaling.
4. Split the dataset into training and testing sets (80:20).
5. Trained Linear Regression, Ridge Regression, and Decision Tree Regressor models.
6. Evaluated model performance using RMSE and R^2 Score.
7. Compared results and selected the best model.

Models Used

- Linear Regression
- Ridge Regression
- Decision Tree Regressor

Results

After training and testing, model performance was compared using RMSE and R^2 Score. The model with the lowest RMSE and highest R^2 Score was selected as the best-performing model.

Conclusion

Feature scaling and model comparison helped improve prediction performance. The selected model demonstrated better generalization and prediction accuracy on the California Housing Dataset.